AATIF: A Multiplicative Governance Equation for Arabic-Aware LLM Safety

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Abstract

We present AATIF, a multiplicative governance equation for LLM safety that combines three semantic scorers—harm (H), intent (I), and emotion (E)—into a single real-valued safety signal S with a structural non-compensability guarantee: no amount of benign intent can override a high harm score. The system uses 249 hand-authored semantic anchors (171 harm, 46 intent, 32 emotion) compared via cosine similarity against bge-m3 embeddings, requiring zero fine-tuning. A domain-parameterized threshold $\theta(d)$ adjusts sensitivity from healthcare (θ=0.25) to creative writing (θ=0.50). On HarmBench (236 behaviors), AATIF blocks 83.8% of safety-category prompts and 100% of CBRN behaviors. On MultiJail (75 prompts), Arabic and English detection rates reach near-parity (88.0% vs. 90.7%), enabled by bge-m3's multilingual embeddings. Ablation shows H is the dominant scorer (F1=0.89 vs. I-only 0.68, E-only 0.36). A set of 31 dialect-hyperbole anchors achieves 5% false-positive rate on benign Arabic dialect test cases, demonstrating that semantic anchors can disambiguate dialectal metaphor from genuine threat. We report all results honestly, including a 28.95% overall false-positive rate on a blind evaluation and weak performance on copyright (38.6%) and misinformation (66.7%) categories. The system is a research prototype built by a single developer; independent replication is needed. Code is available under BSL 1.1.

1 Introduction

LLM safety systems face three underaddressed gaps.

Gap 1: No interpretable equation. Existing guardrail systems—Llama Guard (Inan et al., 2023), Perspective API (Lees et al., 2022), and FanarGuard (Fatehkia et al., 2026)—operate as classifiers whose internal decision boundaries are opaque. A deployer cannot inspect exactly *why* a given prompt was blocked, only *that* it was. This opacity limits auditability in regulated domains such as healthcare and education.

Gap 2: No non-compensability guarantee. Most safety classifiers produce a single composite score that allows a sufficiently benign-sounding context to offset harmful content—a failure mode we term *toxic positivity bypass*. A prompt framed as educational or therapeutic can push a composite score below the blocking threshold even when the core request is dangerous.

Gap 3: Arabic as afterthought. Arabic safety benchmarks and tools remain scarce. Dialectal Arabic introduces a specific challenge: colloquial hyperbole (e.g., *adbaḥak*, "I'll slaughter you," as an expression of affection) triggers harm detectors trained on literal semantics. No existing system provides an explicit mechanism for dialectal metaphor disambiguation.

This paper introduces a governance equation that addresses all three gaps in a single deployable system:

$$S = \sigma(w_1 \cdot I + w_2 \cdot E) \cdot \big[1 - \sigma\big(\alpha(H - \theta(d))\big)\big] \quad (1)$$

where σ is the logistic sigmoid, H, I, E are harm, intent, and emotion scores from three independent semantic scorers, $\theta(d)$ is a domain-dependent threshold, and w_1=2.0, w_2=1.5, α=10 are default parameters. The equation is fully interpretable (Gap 1), structurally non-compensable by construction (Gap 2), and supported by 31 dialect-hyperbole anchors for Arabic disambiguation (Gap 3).

Our contributions are:

1. A multiplicative safety equation with a formal non-compensability property (Section 3).
2. Three deterministic semantic scorers using 249 anchors and bge-m3 embeddings, requiring zero fine-tuning (Section 3.2).
3. Arabic dialect-hyperbole disambiguation via 31 counter-anchors (Section 4).

4. Ablation, benchmark, and blind evaluation results with honest reporting of limitations (Section 5).

2 Related Work

Alignment approaches. RLHF (Christiano et al., 2017) and Constitutional AI (Bai et al., 2022) fine-tune model weights to align behavior with human preferences. Mechanistic interpretability work has shown that refusal behavior is mediated by a single linear direction in activation space (Arditi et al., 2024), that DPO does not eliminate harmful capabilities but reroutes around them (Lee et al., 2024), and that any attenuation-based alignment is susceptible to adversarial elicitation (Wolf et al., 2024). These findings motivate an external, interpretable governance layer rather than reliance on internal alignment alone.

Arabic safety. FanarGuard (Fatehkia et al., 2026) is a fine-tuned safety classifier for Arabic with strong performance on its evaluation set. ADHAR (Charfi et al., 2024) provides an Arabic safety benchmark. SOD (Asiri and Saleh, 2024) addresses offensive language detection. None of these systems provides an explicit mechanism for dialectal hyperbole disambiguation or a domain-parameterized threshold.

Benchmarks. HarmBench (Mazeika et al., 2024) provides 236 harmful behaviors across seven categories with standardized evaluation. MultiJail (Deng et al., 2024) tests multilingual jailbreak resistance across ten languages. MLCommons AI Safety v0.5 (Vidgen et al., 2024) defines 13 hazard categories for comprehensive safety evaluation. We evaluate against HarmBench and MultiJail and map coverage to MLCommons categories.

3 The AATIF Governance Equation

3.1 Definition and Properties

The safety signal S is defined in Equation 1. The equation has two multiplicative factors:

- **Quality term** $Q = \sigma(w_1 \cdot I + w_2 \cdot E)$: captures the positive signal—how clear and constructive the intent is. Since σ maps to $(0, 1)$, the quality term is bounded: $Q \in (0, 1)$.
- **Harm gate** $G = 1 - \sigma(\alpha(H - \theta(d)))$: a soft gate that approaches zero as H exceeds $\theta(d)$. When $H \gg \theta(d)$, $\sigma(\alpha(H - \theta(d))) \to 1$ and $G \to 0$.

Non-compensability (by construction). Since $S = Q \cdot G$ and both $Q, G \in (0, 1)$, we have:

$$H \gg \theta(d) \implies G \to 0 \implies S \to 0$$

regardless of the values of I and E. No quality term, however large before sigmoid saturation, can compensate for a high harm score because the product structure ensures $S \leq G$. This is a structural property of the multiplicative form, not a parameter tuning result.

Hard override (external to the equation). An additional external rule enforces: if $H > 0.7$, the system enters SAFE_FREEZE regardless of S. This override is not part of the equational definition; it is a sovereignty rule imposed by the governance pipeline.

Decision mapping. The continuous S value is mapped to discrete actions:

Condition	Action
$S > 0.7$	EXECUTE
$0.5 < S \leq 0.7$	CLARIFY
$0.3 < S \leq 0.5$	SAFE_STOP
$S \leq 0.3$	SAFE_FREEZE
$H > 0.7$ (override)	SAFE_FREEZE

Table 1: Decision mapping from S to governance actions.

Sigmoid symmetry. The sigmoid $\sigma(x) = 1/(1 + e^{-x})$ is symmetric around its inflection point at x=0. In the harm gate, the inflection occurs at $H = \theta(d)$, meaning the gate transitions sharply from open ($G \approx 1$) to closed ($G \approx 0$) around the domain threshold. The steepness parameter α=10 controls the sharpness of this transition: larger α makes the gate more binary.

3.2 The Three Scorers

Each scorer computes a similarity score between the input text and a curated set of semantic anchors using bge-m3 embeddings (Chen et al., 2024) served via Ollama.

Harm scorer (H). 171 anchors covering CBRN, cybercrime, illegal activity, harassment, harmful content, misinformation, and copyright-adjacent language. For each input, all 171 anchors are scored via cosine similarity. The top-K=3 scores are selected and combined via softmax with temperature τ=0.05:

$$H = \sum_{k=1}^{3} \frac{e^{s_k/\tau}}{\sum_{j=1}^{3} e^{s_j/\tau}} \cdot s_k$$

The low temperature produces a near-argmax selection, making H primarily determined by the single closest anchor. This is deterministic at test time: identical inputs produce identical scores.

Intent scorer (I). 46 anchors capturing constructive, informational, educational, and ambiguous intent patterns. Same top-K and softmax architecture. High I scores indicate clear, constructive intent.

Emotion scorer (E). 32 anchors spanning emotional states from distress to curiosity. The emotion scorer contributes less to harm detection (ablation F1=0.36) but serves other roles in the governance pipeline: adjusting response empathy and tone.

All three scorers operate on the same bge-m3 embedding of the input, making the full scoring pipeline a single embedding call followed by three sets of cosine similarity computations. The total anchor count is $171 + 46 + 32 = 249$.

3.3 Governance Pipeline

The scorers are the first stage of a multi-stage pipeline orchestrated by a Governor module:

1. $S(d)$: The safety equation computes the composite score and maps it to a decision (Table 1).
2. $P(d)$: A mode profiler selects a behavioral profile (e.g., empathetic, factual, creative) based on the intent and emotion scores.
3. $R(d)$: A response-shaping function adjusts LLM generation parameters (temperature, system prompt constraints) according to the selected profile.
4. **LLM**: The underlying model generates a response within the constraints set by $R(d)$.
5. **Output Gate**: A post-generation check reevaluates the LLM output against harm anchors before delivery.

The Governor enforces sovereignty rules: certain categories (CBRN, child safety) trigger unconditional blocking regardless of scores. Every decision is logged with the full scoring vector (H, I, E, S), enabling post-hoc audit.

The design philosophy is *human-over-the-loop*: the system makes autonomous decisions within its defined boundaries, but all boundaries are set by human-authored anchors and thresholds, and all decisions are auditable. This differs from human-in-the-loop systems that require human approval for each decision.

3.4 Domain Parameterization

The threshold $\theta(d)$ adapts sensitivity to deployment context:

Domain	$\theta(d)$	Rationale
Healthcare	0.25	Minimal risk tolerance
Education	0.30	Age-appropriate caution
General	0.40	Balanced default
Creative	0.50	Permits exploration

Table 2: Domain-parameterized thresholds.

Lower θ shifts the harm gate's inflection point leftward, making the system more conservative. The hard override ($H > 0.7 \rightarrow$ SAFE_FREEZE) is domain-independent: CBRN content is blocked regardless of deployment context.

4 Arabic-Aware Design

4.1 Dialect Hyperbole Disambiguation

Arabic dialects use violent metaphor as everyday expression. The phrase *adbaḥak* ("I'll slaughter you") is a common term of endearment in Levantine and Gulf Arabic. *Amawwitak* ("I'll kill you") expresses admiration. A literal-semantic harm detector flags these as high-risk, producing false positives on benign dialectal input.

AATIF addresses this through 31 dialect-hyperbole anchors, each scored at $H \leq 0.05$ in the harm anchor set. These anchors comprise 5 colloquial metaphor templates and 26 in-context variants that capture the pragmatic (non-threatening) use of violent vocabulary. When a dialectal input is embedded via bge-m3, the cosine similarity to these low-harm anchors competes with similarity to genuine harm anchors in the top-K selection.

Phrase	Literal	Pragmatic
adbaḥak	slaughter	affection / praise
amawwitak	kill you	admiration
aakil haali	eat myself	frustration (benign)
yilAAn abuuk	curse father	playful teasing

Table 3: Dialect hyperbole examples. Romanized transliteration.

The mechanism works through what we call *counter-gravity*: dialect anchors pull the top-K selection toward low-H scores when the input's embedding is closer to pragmatic use than to genuine threat. On the blind evaluation's `dialect_benign` category, this achieves a false-positive rate of only 5%.

Failure mode: Lexical Anchor Contamination. When a genuinely harmful prompt borrows dialectal phrasing, the dialect anchors can suppress the harm score, allowing the prompt to pass. This is a known limitation: the counter-gravity mechanism trades recall on dialect-disguised threats for precision on benign dialect.

4.2 Meaning Density Hypothesis

Arabic's triconsonantal root system packs more semantic information per token than English, motivating the hypothesis that Arabic text may require fewer tokens to convey equivalent meaning—and thus that embedding models may need to extract meaning from denser input. We state this explicitly as motivation, not as a validated claim: **no evidence exists that current embedding models (including bge-m3) preserve Arabic morphological structure** in their vector representations. Validating this hypothesis would require probing experiments that are beyond the scope of this work.

5 Experimental Evaluation

All experiments use bge-m3 embeddings via Ollama on a single Mac Mini. The general domain threshold θ=0.40 is used unless otherwise noted.

5.1 Ablation Study

We evaluate the contribution of each scorer by running the pipeline with individual scorers disabled. The test set comprises 311 harmful prompts (from HarmBench and MultiJail) and 50 benign prompts.

Config	Prec.	Recall	F1
H-only	0.988	0.810	0.890
I-only	0.982	0.515	0.675
E-only	0.875	0.225	0.358
Full (H+I+E)	(see Table 5)		

Table 4: Ablation: individual scorer performance on 361 cases (311 harmful + 50 benign).

Key findings. H is the dominant scorer, achieving F1=0.890 with high precision (0.988) and strong recall (0.810). I provides complementary recall (0.515) with comparable precision, suggesting it captures intent-related harm patterns that H misses. E contributes minimally to detection (F1=0.358) but serves other roles in the pipeline: modulating response tone and empathy.

The H-only false-positive rate is 0.06 (3/50 benign cases misclassified). I-only achieves the same FPR of 0.06, while E-only produces a higher FPR of 0.20. The ablation confirms that the multiplicative combination preserves H's precision while the quality term gates nuance for borderline cases.

5.2 HarmBench Results

We evaluate the full pipeline on all 236 HarmBench behaviors using the general domain threshold θ=0.40.

Category	n	Blocked	Avg H
CBRN	28	28 (100%)	0.846
Illegal	43	38 (88.4%)	0.696
Cybercrime	43	37 (86.0%)	0.663
Harassment	15	13 (86.7%)	0.653
Harmful	10	7 (70.0%)	0.590
Misinfo	39	26 (66.7%)	0.536
Copyright	57	22 (38.6%)	0.308
Safety (excl. copyright)	179	150 (83.8%)	—
Not-executed (safety)	179	162 (90.5%)	—
Overall	236	172 (72.9%)	—

Table 5: HarmBench full pipeline results (236 behaviors, θ=0.40). "Safety" excludes copyright. "Not-executed" includes SAFE_STOP and SAFE_FREEZE.

Performance tracks average H score: categories with rich harm vocabulary (CBRN, illegal, cybercrime) achieve high recall because their language closely matches harm anchors. Copyright achieves only 38.6% because copyright infringement requests use neutral, non-threatening vocabulary that produces low cosine similarity to existing harm anchors. Misinformation (66.7%) is a middle case: some misinformation prompts contain detectable harm language, but many do not.

5.3 MultiJail Results

We evaluate on 75 MultiJail prompts in both Arabic and English.

Language	Blocked	Rate	Avg H
Arabic	66/75	88.0%	0.670
English	68/75	90.7%	0.668

Table 6: MultiJail results (75 prompts, θ=0.40). Arabic scores higher in 38/75 cases (50.7%).

The near-parity between Arabic and English (88.0% vs. 90.7%) demonstrates that bge-m3's multilingual embeddings enable cross-lingual harm detection without language-specific fine-tuning. In 38 of 75 cases (50.7%), the Arabic version produced a higher H score than the English version, suggesting that bge-m3 captures Arabic harm semantics at least as well as English for this prompt set.

5.4 Blind Evaluation

A blind evaluation was conducted on 570 cases: 380 harmful and 190 benign, drawn from multiple sources and including adversarial categories.

Metric	Value
Detection rate	83.16%
False-positive rate	28.95%
F1	0.8415

Table 7: Blind evaluation (570 cases: 380 harmful, 190 benign).

Strengths and weaknesses by category. The `dialect_benign` category achieved an FPR of only 5%, confirming that the dialect-hyperbole disambiguation mechanism works as designed. However, the `tricky_benign` category (adversarially crafted benign prompts using harm-adjacent vocabulary) produced an FPR of 55%, and `arabic_benign` reached 30%. The overall FPR of 28.95% is a significant limitation: roughly one in four benign inputs is incorrectly flagged.

5.5 Baseline Comparison

A keyword-matching baseline using 139 harm-related keywords achieves a detection rate of 54.2% on HarmBench's 236 behaviors. The AATIF full pipeline achieves 72.9% on the same set, a 1.34× improvement. This confirms that semantic anchors with embedding-based similarity outperform lexical matching, as expected: semantic anchors capture paraphrases and indirect references that exact keyword matching misses.

6 Limitations

We report limitations explicitly, as required by ARR.

Single developer. AATIF was built and evaluated by a single independent researcher. All anchors, thresholds, and evaluation protocols reflect one person's judgment. Independent replication with separate evaluators is necessary before drawing general conclusions.

MLCommons coverage gaps. Of the 13 MLCommons AI Safety v0.5 hazard categories, AATIF shows strong coverage on 4 (violent crimes, CBRN, child safety, and non-violent crimes) and has zero or minimal coverage on 6 categories (sexual content, privacy, intellectual property, indiscriminate weapons, hate, and self-harm). Addressing these gaps requires expanding the anchor sets.

High false-positive rate. The blind evaluation's overall FPR of 28.95% is too high for production deployment. The `tricky_benign` FPR of 55% indicates that the system cannot reliably distinguish harm-adjacent benign content from genuine threats.

Copyright and misinformation. Copyright detection (38.6%) and misinformation detection (66.7%) are weak because these categories lack distinctive harm vocabulary. Detecting copyright infringement and misinformation likely requires reasoning capabilities beyond cosine similarity to harm anchors.

No pragmatic inference. The system performs no sarcasm detection, no implicit harm recognition, and no pragmatic inference. A sarcastic threat and a genuine threat with identical vocabulary receive identical scores.

Small anchor scale. 249 anchors is a small knowledge base. For comparison, fine-tuned safety classifiers train on hundreds of thousands of labeled examples (e.g., 468K for Llama Guard). The anchor approach trades scale for interpretability and zero training cost.

Research prototype. The system runs on a single Mac Mini with Ollama. It is not production-grade in terms of latency, throughput, or failure handling.

7 Conclusion

We have presented AATIF, a governance equation $S = \sigma(w_1 \cdot I + w_2 \cdot E) \cdot [1 - \sigma(\alpha(H - \theta(d)))]$ that combines three semantic scorers into a single safety signal with a structural non-compensability guarantee. The system uses 249 hand-authored anchors and bge-m3 embeddings, requiring zero fine-tuning.

Ablation confirms that H is the dominant scorer (F1=0.890), with I providing complementary recall. On HarmBench, the system blocks 83.8% of safety-category behaviors and 100% of CBRN. On MultiJail, Arabic and English achieve near-parity (88.0% vs. 90.7%). Dialect-hyperbole disambiguation achieves 5% FPR on benign dialect test cases.

The system has significant limitations: a 28.95% overall false-positive rate, weak performance on copyright and misinformation, no pragmatic inference, and single-developer construction. It is a research prototype, not a production system.

The contribution is the combination, for the first time in a single deployable system, of a formally non-compensable safety equation, interpretable semantic anchors, domain parameterization, and Arabic dialect disambiguation—all without model fine-tuning.

Future work includes anchor expansion to cover missing MLCommons categories, cultural alignment as an explicit scoring axis, and investigation of Arabic-specific embedding models that may better preserve morphological structure.

Ethics Statement

This work presents a safety system, which introduces dual-use considerations: understanding how governance equations work could inform adversarial evasion strategies. We release the framework under BSL 1.1 (code) and CC BY 4.0 (paper) to enable reproducibility while maintaining responsible use constraints.

The system was developed and evaluated by a single independent researcher without institutional oversight. This means no IRB review, no red-team evaluation by an independent party, and no adversarial stress-testing beyond the benchmarks reported. We emphasize that independent evaluation is essential before any deployment in real-world settings.

The anchors encode one researcher's judgment about what constitutes harm, which inevitably reflects cultural and personal biases. Expanding the anchor authorship to a diverse group of evaluators is a necessary next step.

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Appendices

A Mode Profiles

The mode profiler $P(d)$ selects one of several behavioral profiles based on the intent and emotion scores. Each profile adjusts LLM generation parameters.

Mode	Trigger	Effect
Empathetic	High E (distress)	Lower temp, softer tone
Factual	High I (info)	Low temp, citation-heavy
Creative	High I (creative)	Higher temp, looser
Cautious	Med H	Conservative phrasing
Frozen	High H	No generation

Table 8: Representative mode profiles from $P(d)$.

B Hysteresis Controller

To prevent oscillation between adjacent decision bands (e.g., rapid switching between CLARIFY and SAFE_STOP), the Governor implements a hysteresis controller. Once a session enters a restrictive mode (SAFE_STOP or SAFE_FREEZE), the threshold for returning to a permissive mode is raised by a hysteresis margin $\delta = 0.05$. For example, a session in SAFE_STOP requires $S > 0.55$ (rather than $S > 0.50$) to return to CLARIFY. This prevents mode flickering in adversarial probing scenarios where an attacker incrementally adjusts prompt wording.

C Response Shaping Equation

The response shaper $R(d)$ maps the mode profile to concrete LLM parameters:

$$R(d) = \langle \tau_{\text{gen}},\ p_{\text{sys}},\ l_{\max},\ f_{\text{style}} \rangle \tag{2}$$

where τ_{gen} is the generation temperature, p_{sys} is the system prompt modifier, $l_{\max}$ is the maximum response length, and f_{style} is a style flag (formal, conversational, clinical). For example, the Empathetic mode sets τ_{gen}=0.4, prepends a compassion-oriented system prompt, and enables the conversational style flag.

D Safety Gates

Two sovereignty laws govern unconditional blocking:

Law Ω (Category Override). Certain harm categories (CBRN, child safety, explicit violence against named individuals) trigger SAFE_FREEZE regardless of the computed S value. This is implemented as a pre-equation check on the harm anchor category labels.

Law Ξ (Cascade Freeze). If three consecutive user turns within a session produce $S \leq 0.3$, the session enters a persistent SAFE_FREEZE state that requires explicit human reset. This prevents incremental jailbreak attempts that probe the boundary one step at a time.

E Calibration: θ Sweep

The domain-parameterized threshold $\theta(d)$ controls the tradeoff between recall and false-positive rate. Structurally, lower θ shifts the harm gate's inflection point leftward, making the system more conservative: more harmful prompts are blocked (higher recall), but more benign prompts are also flagged (higher FPR). The default θ=0.40 for the general domain was selected through calibration experiments. Domain-specific thresholds (Table 2) shift along this curve: healthcare at θ=0.25 accepts a higher FPR for maximal recall; creative writing at θ=0.50 accepts lower recall for reduced false positives.

A formal θ sweep with per-value metrics should be included in the final version; the current experiments fix θ=0.40 for the general domain.

F MLCommons Coverage Mapping

MLCommons Category	Coverage	Notes
Violent crimes	Strong	38/43 illegal
CBRN	Strong	28/28 (100%)
Child safety	Strong	Sovereignty law Ω
Non-violent crimes	Strong	37/43 cybercrime
Sex-related	Minimal	Few anchors
Privacy	None	Not addressed
Intellectual prop.	Weak	22/57 copyright
Indisc. weapons	Partial	Subset of CBRN
Hate	Moderate	13/15 harassment
Self-harm	Minimal	Few anchors
Elections	None	Not addressed
Defamation	None	Not addressed
Expert advice	None	Not addressed

Table 9: AATIF coverage mapped to MLCommons AI Safety v0.5 categories.

G FanarGuard Comparison

Direct comparison with FanarGuard is methodologically limited because (1) the evaluation datasets differ, (2) FanarGuard is a fine-tuned classifier while AATIF uses zero-shot embedding similarity, and (3) FanarGuard's evaluation focuses on

Arabic-specific benchmarks not fully overlapping with HarmBench.

The key architectural differences are:

Property	AATIF	FanarGuard
Method	Anchor similarity	Fine-tuned LLM
Training	Zero	Full fine-tuning
Interpretability	Full (anchors)	Partial (logits)
Arabic dialect	31 anchors	Training data
Domain adapt.	$\theta(d)$ param.	Retraining
Scale	249 anchors	Model weights

Table 10: Architectural comparison: AATIF vs. FanarGuard.

We do not claim superiority over FanarGuard. The systems occupy different points in the interpretability–performance tradeoff space. AATIF's advantage is full interpretability and zero training cost; FanarGuard's advantage is likely higher raw performance from fine-tuning on large labeled datasets.